ValPay Portfolio 1: Database Design

MIT 8103 Advanced Database Systems · Evidence

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Compiled from `portfolio-1-database-design/evidence`: 5 screenshots in `screenshots/` and script output in `script-output/` (`schema_dump.txt`). Text output is reproduced verbatim.

1. ENTITY RELATIONSHIP DIAGRAM

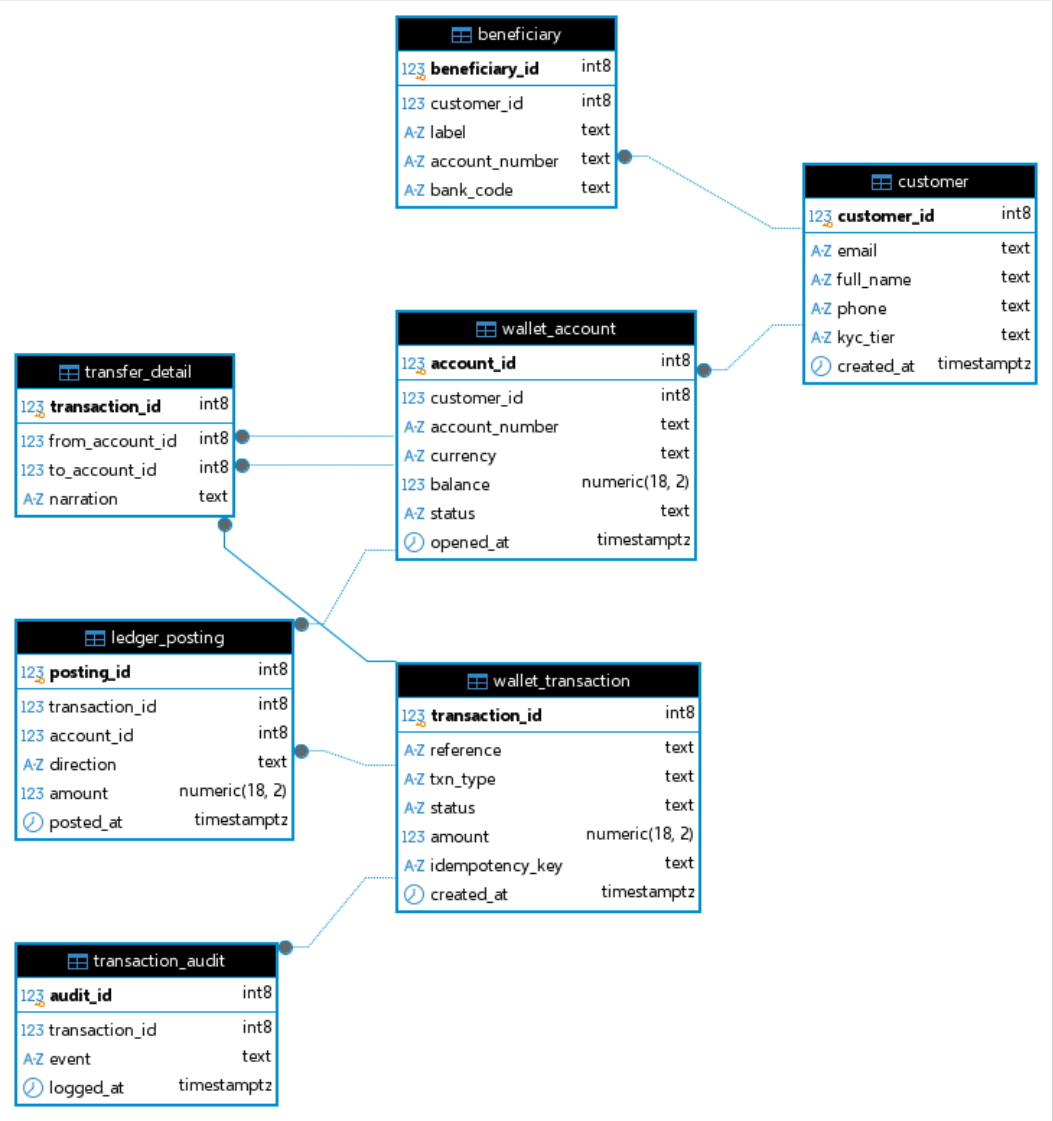


Figure 1. *ValPay ER diagram exported from DBeaver: seven tables with columns, data types, primary keys and relationships.*

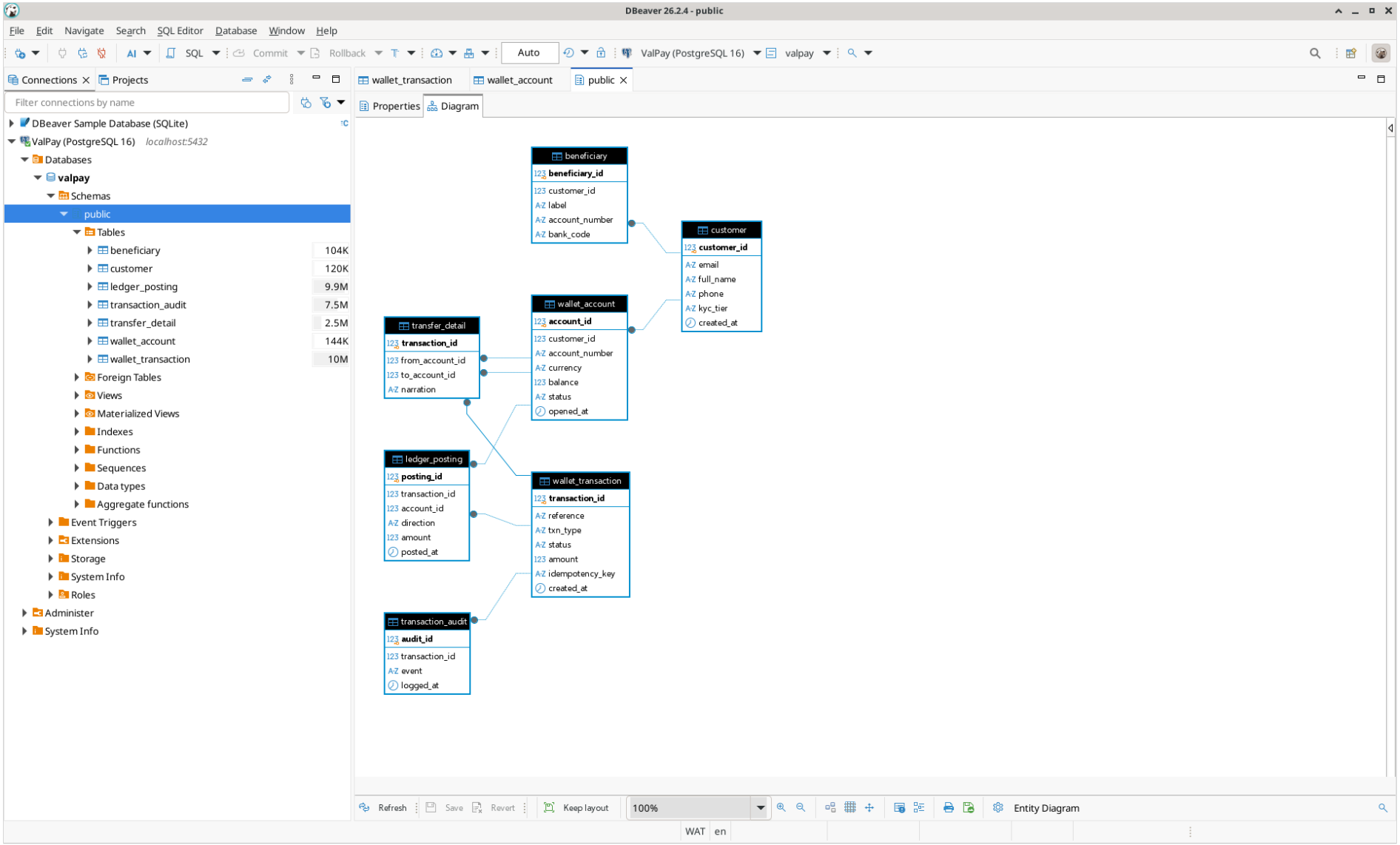


Figure 2. *DBeaver Diagram tab for the public schema of the valpay database.*

2. TABLES AND COLUMNS

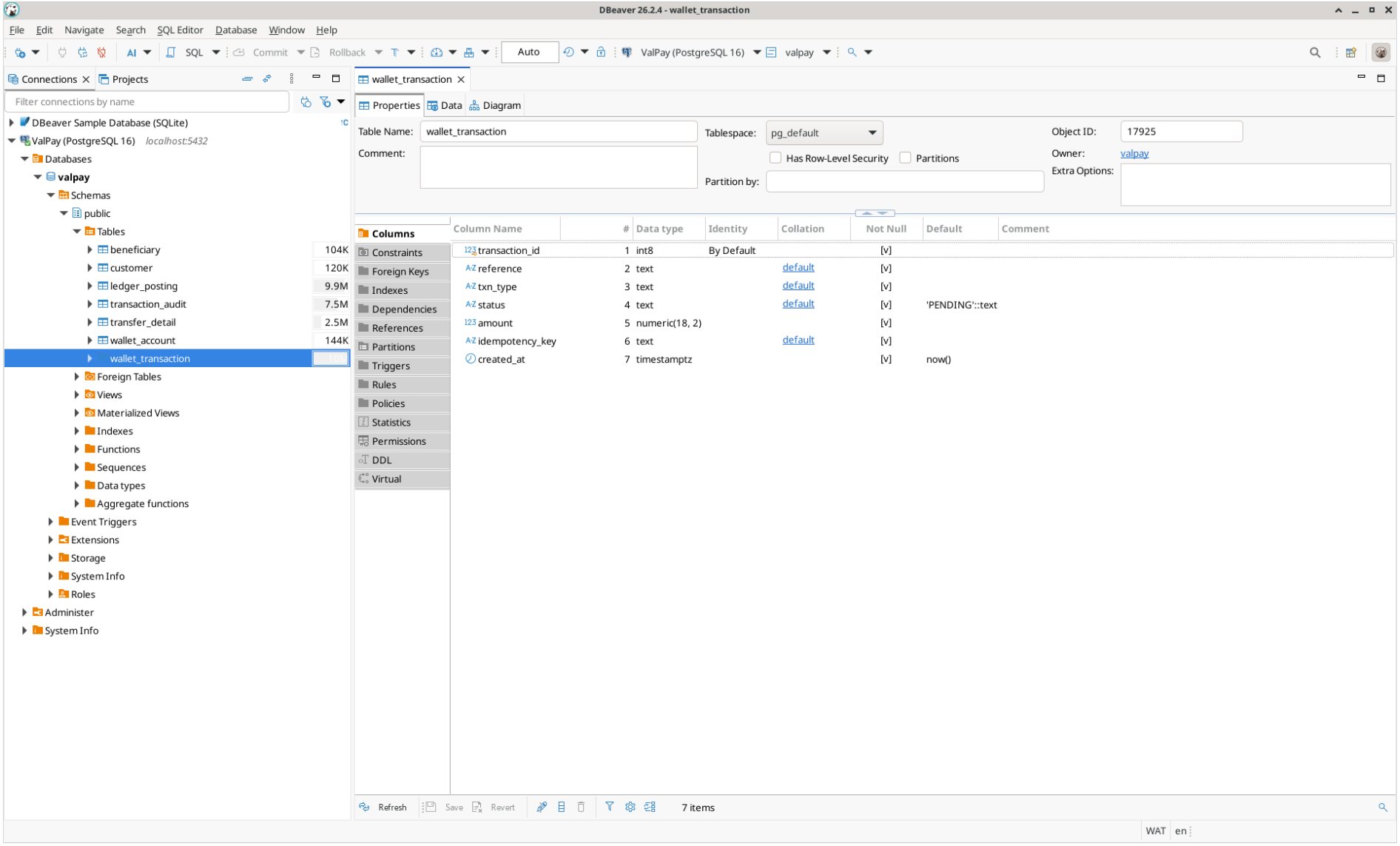


Figure 3. *DBeaver table editor: wallet_transaction columns, data types, defaults and NOT NULL settings.*

3. CONSTRAINTS AND INDEXES

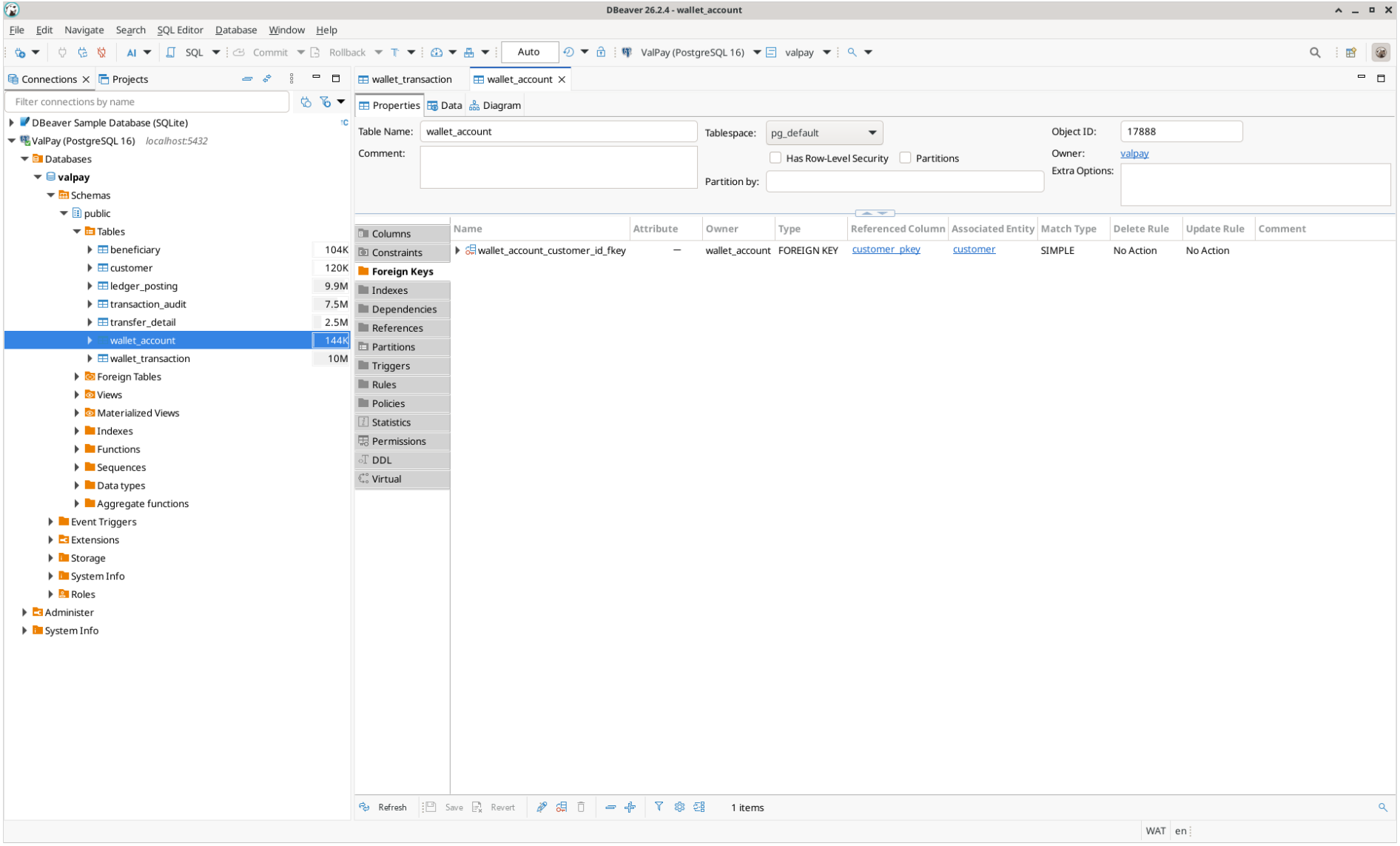


Figure 4. *DBeaver: wallet_account constraints, including the foreign key to customer.*

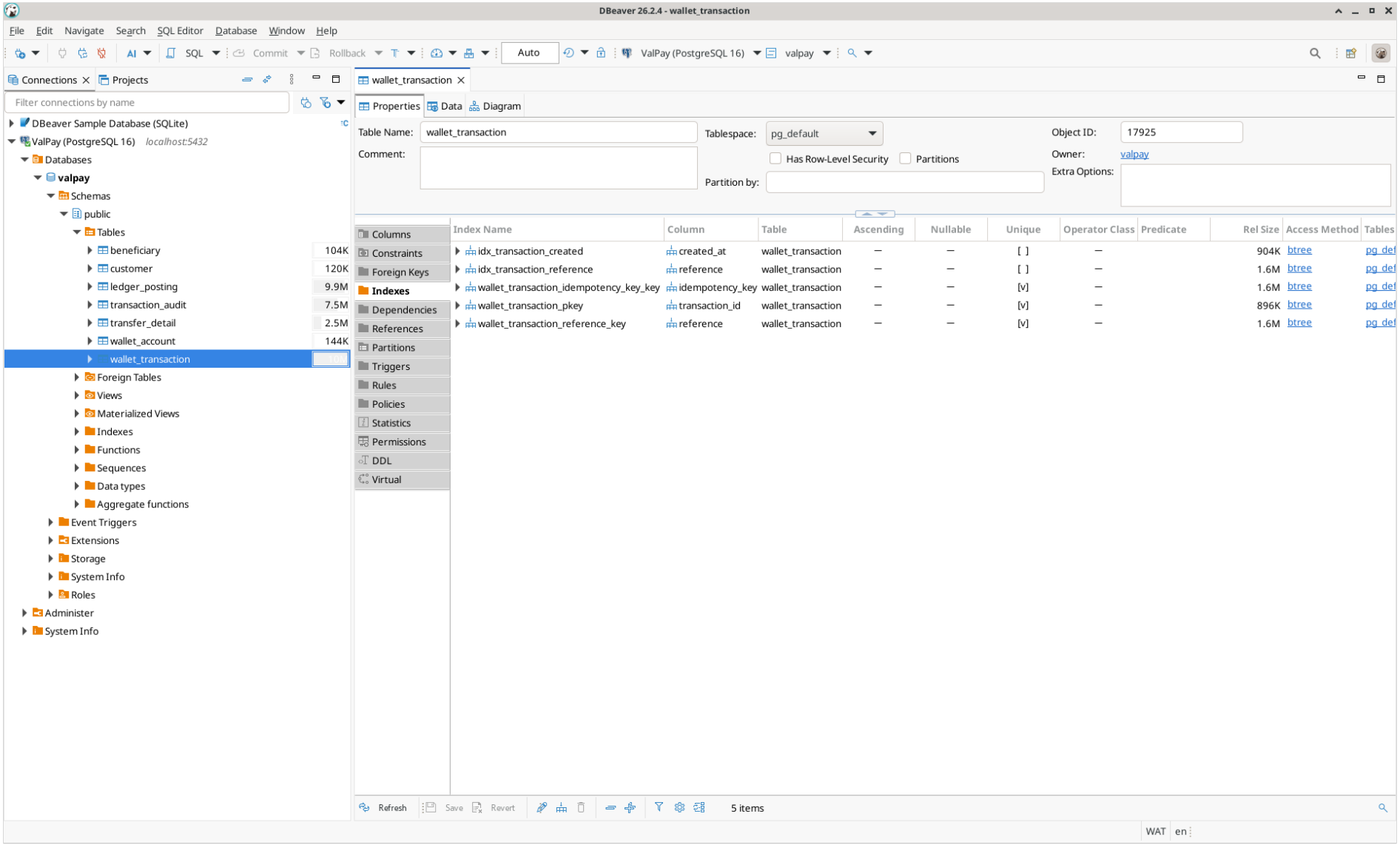


Figure 5. *DBeaver: wallet_transaction indexes, including idx_transaction_reference (text_pattern_ops).*

4. SCHEMA DUMP (PG_DUMP --SCHEMA-ONLY)

Source: evidence/script-output/schema_dump.txt

```
--
-- PostgreSQL database dump
--

\restrict 7Sex6NwEH0ZqiHoYpplLzDFS8IKRwaGsAZnyYLAhijxS6jA8fl40KK4hMGoZeip

-- Dumped from database version 16.15
-- Dumped by pg_dump version 16.15

SET statement_timeout = 0;
SET lock_timeout = 0;
SET idle_in_transaction_session_timeout = 0;
SET client_encoding = 'UTF8';
SET standard_conforming_strings = on;
SELECT pg_catalog.set_config('search_path', '', false);
SET check_function_bodies = false;
SET xmloption = content;
SET client_min_messages = warning;
SET row_security = off;

--
-- Name: public; Type: SCHEMA; Schema: -; Owner: -
--

-- *not* creating schema, since initdb creates it


--
-- Name: SCHEMA public; Type: COMMENT; Schema: -; Owner: -
--

COMMENT ON SCHEMA public IS '';


SET default_tablespace = '';

SET default_table_access_method = heap;

--
-- Name: beneficiary; Type: TABLE; Schema: public; Owner: -
--

CREATE TABLE public.beneficiary (
    beneficiary_id bigint NOT NULL,
    customer_id bigint NOT NULL,
    label text NOT NULL,
    account_number text NOT NULL,
    bank_code text NOT NULL
);


--
-- Name: beneficiary_beneficiary_id_seq; Type: SEQUENCE; Schema: public; Owner: -
--

ALTER TABLE public.beneficiary ALTER COLUMN beneficiary_id ADD GENERATED BY DEFAULT AS IDENTITY (
    SEQUENCE NAME public.beneficiary_beneficiary_id_seq
    START WITH 1
    INCREMENT BY 1
    NO MINVALUE
    NO MAXVALUE
    CACHE 1
);


--
-- Name: customer; Type: TABLE; Schema: public; Owner: -
--
```

```
CREATE TABLE public.customer (
    customer_id bigint NOT NULL,
    email text NOT NULL,
    full_name text NOT NULL,
    phone text NOT NULL,
    kyc_tier text DEFAULT 'TIER1'::text NOT NULL,
    created_at timestamp with time zone DEFAULT now() NOT NULL,
    CONSTRAINT customer_kyc_tier_check CHECK ((kyc_tier = ANY (ARRAY['TIER1'::text, 'TIER2'::text,
'TIER3'::text])))
);


--
-- Name: customer_customer_id_seq; Type: SEQUENCE; Schema: public; Owner: -
--

ALTER TABLE public.customer ALTER COLUMN customer_id ADD GENERATED BY DEFAULT AS IDENTITY (
    SEQUENCE NAME public.customer_customer_id_seq
    START WITH 1
    INCREMENT BY 1
    NO MINVALUE
    NO MAXVALUE
    CACHE 1
);


--
-- Name: ledger_posting; Type: TABLE; Schema: public; Owner: -
--

CREATE TABLE public.ledger_posting (
    posting_id bigint NOT NULL,
    transaction_id bigint NOT NULL,
    account_id bigint NOT NULL,
    direction text NOT NULL,
    amount numeric(18,2) NOT NULL,
    posted_at timestamp with time zone DEFAULT now() NOT NULL,
    CONSTRAINT ledger_posting_amount_check CHECK ((amount > (0)::numeric)),
    CONSTRAINT ledger_posting_direction_check CHECK ((direction = ANY (ARRAY['DEBIT'::text,
'CREDIT'::text])))
);


--
-- Name: ledger_posting_posting_id_seq; Type: SEQUENCE; Schema: public; Owner: -
--

ALTER TABLE public.ledger_posting ALTER COLUMN posting_id ADD GENERATED BY DEFAULT AS IDENTITY (
    SEQUENCE NAME public.ledger_posting_posting_id_seq
    START WITH 1
    INCREMENT BY 1
    NO MINVALUE
    NO MAXVALUE
    CACHE 1
);


--
-- Name: transaction_audit; Type: TABLE; Schema: public; Owner: -
--

CREATE TABLE public.transaction_audit (
    audit_id bigint NOT NULL,
    transaction_id bigint NOT NULL,
    event text NOT NULL,
    logged_at timestamp with time zone DEFAULT now() NOT NULL
);


--
-- Name: transaction_audit_audit_id_seq; Type: SEQUENCE; Schema: public; Owner: -
--
```

```
ALTER TABLE public.transaction_audit ALTER COLUMN audit_id ADD GENERATED BY DEFAULT AS IDENTITY (
    SEQUENCE NAME public.transaction_audit_audit_id_seq
    START WITH 1
    INCREMENT BY 1
    NO MINVALUE
    NO MAXVALUE
    CACHE 1
);


--
-- Name: transfer_detail; Type: TABLE; Schema: public; Owner: -
--

CREATE TABLE public.transfer_detail (
    transaction_id bigint NOT NULL,
    from_account_id bigint NOT NULL,
    to_account_id bigint NOT NULL,
    narration text,
    CONSTRAINT transfer_detail_check CHECK ((from_account_id <> to_account_id))
);


--
-- Name: wallet_account; Type: TABLE; Schema: public; Owner: -
--

CREATE TABLE public.wallet_account (
    account_id bigint NOT NULL,
    customer_id bigint NOT NULL,
    account_number text NOT NULL,
    currency text DEFAULT 'NGN'::text NOT NULL,
    balance numeric(18,2) DEFAULT 0 NOT NULL,
    status text DEFAULT 'ACTIVE'::text NOT NULL,
    opened_at timestamp with time zone DEFAULT now() NOT NULL,
    CONSTRAINT wallet_account_balance_check CHECK ((balance >= (0)::numeric)),
    CONSTRAINT wallet_account_currency_check CHECK ((currency = 'NGN'::text)),
    CONSTRAINT wallet_account_status_check CHECK ((status = ANY (ARRAY['ACTIVE'::text, 'FROZEN'::text,
'CLOSED'::text])))
);


--
-- Name: wallet_account_account_id_seq; Type: SEQUENCE; Schema: public; Owner: -
--

ALTER TABLE public.wallet_account ALTER COLUMN account_id ADD GENERATED BY DEFAULT AS IDENTITY (
    SEQUENCE NAME public.wallet_account_account_id_seq
    START WITH 1
    INCREMENT BY 1
    NO MINVALUE
    NO MAXVALUE
    CACHE 1
);


--
-- Name: wallet_transaction; Type: TABLE; Schema: public; Owner: -
--

CREATE TABLE public.wallet_transaction (
    transaction_id bigint NOT NULL,
    reference text NOT NULL,
    txn_type text NOT NULL,
    status text DEFAULT 'PENDING'::text NOT NULL,
    amount numeric(18,2) NOT NULL,
    idempotency_key text NOT NULL,
    created_at timestamp with time zone DEFAULT now() NOT NULL,
    CONSTRAINT wallet_transaction_amount_check CHECK ((amount > (0)::numeric)),
    CONSTRAINT wallet_transaction_status_check CHECK ((status = ANY (ARRAY['PENDING'::text,
'SUCCESS'::text, 'FAILED'::text, 'REVERSED'::text]))),
    CONSTRAINT wallet_transaction_txn_type_check CHECK ((txn_type = ANY (ARRAY['TRANSFER'::text,
```

```
'TOPUP'::text, 'BILL_PAY'::text, 'REVERSAL'::text])))
);


--
-- Name: wallet_transaction_transaction_id_seq; Type: SEQUENCE; Schema: public; Owner: -
--

ALTER TABLE public.wallet_transaction ALTER COLUMN transaction_id ADD GENERATED BY DEFAULT AS IDENTITY (
    SEQUENCE NAME public.wallet_transaction_transaction_id_seq
    START WITH 1
    INCREMENT BY 1
    NO MINVALUE
    NO MAXVALUE
    CACHE 1
);


--
-- Name: beneficiary beneficiary_customer_id_account_number_bank_code_key; Type: CONSTRAINT; Schema:
public; Owner: -
--

ALTER TABLE ONLY public.beneficiary
    ADD CONSTRAINT beneficiary_customer_id_account_number_bank_code_key UNIQUE (customer_id,
account_number, bank_code);


--
-- Name: beneficiary beneficiary_pkey; Type: CONSTRAINT; Schema: public; Owner: -
--

ALTER TABLE ONLY public.beneficiary
    ADD CONSTRAINT beneficiary_pkey PRIMARY KEY (beneficiary_id);


--
-- Name: customer customer_email_key; Type: CONSTRAINT; Schema: public; Owner: -
--

ALTER TABLE ONLY public.customer
    ADD CONSTRAINT customer_email_key UNIQUE (email);


--
-- Name: customer customer_phone_key; Type: CONSTRAINT; Schema: public; Owner: -
--

ALTER TABLE ONLY public.customer
    ADD CONSTRAINT customer_phone_key UNIQUE (phone);


--
-- Name: customer customer_pkey; Type: CONSTRAINT; Schema: public; Owner: -
--

ALTER TABLE ONLY public.customer
    ADD CONSTRAINT customer_pkey PRIMARY KEY (customer_id);


--
-- Name: ledger_posting ledger_posting_pkey; Type: CONSTRAINT; Schema: public; Owner: -
--

ALTER TABLE ONLY public.ledger_posting
    ADD CONSTRAINT ledger_posting_pkey PRIMARY KEY (posting_id);


--
-- Name: transaction_audit transaction_audit_pkey; Type: CONSTRAINT; Schema: public; Owner: -
--

ALTER TABLE ONLY public.transaction_audit
```

```
    ADD CONSTRAINT transaction_audit_pkey PRIMARY KEY (audit_id);


--
-- Name: transfer_detail transfer_detail_pkey; Type: CONSTRAINT; Schema: public; Owner: -
--

ALTER TABLE ONLY public.transfer_detail
    ADD CONSTRAINT transfer_detail_pkey PRIMARY KEY (transaction_id);


--
-- Name: wallet_account wallet_account_account_number_key; Type: CONSTRAINT; Schema: public; Owner: -
--

ALTER TABLE ONLY public.wallet_account
    ADD CONSTRAINT wallet_account_account_number_key UNIQUE (account_number);


--
-- Name: wallet_account wallet_account_pkey; Type: CONSTRAINT; Schema: public; Owner: -
--

ALTER TABLE ONLY public.wallet_account
    ADD CONSTRAINT wallet_account_pkey PRIMARY KEY (account_id);


--
-- Name: wallet_transaction wallet_transaction_idempotency_key_key; Type: CONSTRAINT; Schema: public;
Owner: -
--

ALTER TABLE ONLY public.wallet_transaction
    ADD CONSTRAINT wallet_transaction_idempotency_key_key UNIQUE (idempotency_key);


--
-- Name: wallet_transaction wallet_transaction_pkey; Type: CONSTRAINT; Schema: public; Owner: -
--

ALTER TABLE ONLY public.wallet_transaction
    ADD CONSTRAINT wallet_transaction_pkey PRIMARY KEY (transaction_id);


--
-- Name: wallet_transaction wallet_transaction_reference_key; Type: CONSTRAINT; Schema: public; Owner: -
--

ALTER TABLE ONLY public.wallet_transaction
    ADD CONSTRAINT wallet_transaction_reference_key UNIQUE (reference);


--
-- Name: idx_ledger_account; Type: INDEX; Schema: public; Owner: -
--

CREATE INDEX idx_ledger_account ON public.ledger_posting USING btree (account_id);


--
-- Name: idx_ledger_txn; Type: INDEX; Schema: public; Owner: -
--

CREATE INDEX idx_ledger_txn ON public.ledger_posting USING btree (transaction_id);


--
-- Name: idx_transaction_created; Type: INDEX; Schema: public; Owner: -
--

CREATE INDEX idx_transaction_created ON public.wallet_transaction USING btree (created_at);
```

```
--
-- Name: idx_transaction_reference; Type: INDEX; Schema: public; Owner: -
--

CREATE INDEX idx_transaction_reference ON public.wallet_transaction USING btree (reference
text_pattern_ops);


--
-- Name: idx_wallet_customer; Type: INDEX; Schema: public; Owner: -
--

CREATE INDEX idx_wallet_customer ON public.wallet_account USING btree (customer_id);


--
-- Name: beneficiary beneficiary_customer_id_fkey; Type: FK CONSTRAINT; Schema: public; Owner: -
--

ALTER TABLE ONLY public.beneficiary
    ADD CONSTRAINT beneficiary_customer_id_fkey FOREIGN KEY (customer_id) REFERENCES
public.customer(customer_id) ON DELETE CASCADE;


--
-- Name: ledger_posting ledger_posting_account_id_fkey; Type: FK CONSTRAINT; Schema: public; Owner: -
--

ALTER TABLE ONLY public.ledger_posting
    ADD CONSTRAINT ledger_posting_account_id_fkey FOREIGN KEY (account_id) REFERENCES
public.wallet_account(account_id);


--
-- Name: ledger_posting ledger_posting_transaction_id_fkey; Type: FK CONSTRAINT; Schema: public; Owner: -
--

ALTER TABLE ONLY public.ledger_posting
    ADD CONSTRAINT ledger_posting_transaction_id_fkey FOREIGN KEY (transaction_id) REFERENCES
public.wallet_transaction(transaction_id);


--
-- Name: transaction_audit transaction_audit_transaction_id_fkey; Type: FK CONSTRAINT; Schema: public;
Owner: -
--

ALTER TABLE ONLY public.transaction_audit
    ADD CONSTRAINT transaction_audit_transaction_id_fkey FOREIGN KEY (transaction_id) REFERENCES
public.wallet_transaction(transaction_id);


--
-- Name: transfer_detail transfer_detail_from_account_id_fkey; Type: FK CONSTRAINT; Schema: public;
Owner: -
--

ALTER TABLE ONLY public.transfer_detail
    ADD CONSTRAINT transfer_detail_from_account_id_fkey FOREIGN KEY (from_account_id) REFERENCES
public.wallet_account(account_id);


--
-- Name: transfer_detail transfer_detail_to_account_id_fkey; Type: FK CONSTRAINT; Schema: public; Owner:
-
--

ALTER TABLE ONLY public.transfer_detail
    ADD CONSTRAINT transfer_detail_to_account_id_fkey FOREIGN KEY (to_account_id) REFERENCES
public.wallet_account(account_id);


--
```

```
-- Name: transfer_detail transfer_detail_transaction_id_fkey; Type: FK CONSTRAINT; Schema: public; Owner:
-
--

ALTER TABLE ONLY public.transfer_detail
    ADD CONSTRAINT transfer_detail_transaction_id_fkey FOREIGN KEY (transaction_id) REFERENCES
public.wallet_transaction(transaction_id);


--
-- Name: wallet_account wallet_account_customer_id_fkey; Type: FK CONSTRAINT; Schema: public; Owner: -
--

ALTER TABLE ONLY public.wallet_account
    ADD CONSTRAINT wallet_account_customer_id_fkey FOREIGN KEY (customer_id) REFERENCES
public.customer(customer_id);


--
-- PostgreSQL database dump complete
--

\unrestrict 7Sex6NwEH0ZqiHoYpplLzDFS8IKRwaGsAZnyYLAhijxS6jA8fl40KK4hMGoZeip
```